# Joel Julin

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## EDUCATION

**Carnegie Mellon University** **Pittsburgh, PA, USA**
*Ph.D. in Robotics (Advisor: László Jeni)* *Aug. 2025 – present*
*M.S. in Robotics (GPA: 4.19)* *Aug. 2023 – Aug. 2025*
*Thesis:* Towards Geometric Reasoning for Dynamic and Unconstrained 3D Scenes

**University of Pittsburgh** **Pittsburgh, PA, USA**
*B.S. in Computer Science* *Aug. 2020 – Dec. 2022*

## RESEARCH EXPERIENCES

**Research Intern**
*Fujitsu Research of America, PA, USA* *Jan. 2026 – Apr. 2026*
- Collaborated with Mose Sakashita and Koichiro Niinuma on a project for improving the quality of VGGT generated scenes with 3D Gaussian Splatting.
- Prototyped and evaluated a hybrid VGGT + 3DGS pipeline, benchmarking reconstruction quality against existing scene-generation baselines.

**Research Assistant**
*Carnegie Mellon University, PA, USA* *Aug. 2023 – present*
- Worked with László Jeni on neural representations for dynamic 3D scenes.
- Helped lead multiple industry-sponsored research collaborations with Apple, Amazon, SPREEAI, and Fujitsu, spanning 3D computer vision, neural rendering, and spatial reasoning.
- Developed DyLiN [7] for non-rigid dynamic reconstruction, CoGS [6] for real-time controllable Gaussian Splatting, and GSLK [5] for analytically derived scene flow under minimal camera motion.
- Developed distractor-free 4D Gaussian Splatting [1] (WACV 2027) and ANCHOR [2] (NeurIPS 2026), an audio-visually grounded chain-of-thought benchmark.
- Built the first (at the time) interactive dynamic 3D Gaussian Splatting VR viewer.

**Research Intern**
*Carnegie Mellon University, PA, USA* *May 2022 – Aug. 2023*
- RISS intern with László Jeni, researching automated content editing in NeRFs, resulted in [9].

**Research Intern**
*Carnegie Mellon University, PA, USA* *May 2021 – May 2022*
- Researched spatial vs. channel mixing in CNNs with Simon Lucey, led to [4].

## RESEARCH

[1] **J. Julin**, L. Xie, A. Bal, M. Sakashita, K. Niinuma, L. Jeni. *Reconstruct What You Can: Distractor-free 4D Gaussian Splatting*. **WACV** 2027
[2] **J. Julin***, S. Kundu*, L. Dahiya, G. Wei, L. Xie, J. Yi, X. Zhang, H. Yang, L. Jeni. *ANCHOR: Audio-Visually Grounded Chain-of-Thought Reasoning Benchmark*. **NeurIPS** 2026 (Evaluations & Datasets). *Equal contribution
[3] Y. Li, C. Zheng, Y. He, **J. Julin**, R. Ichikari, S. Ogiso, S. Nakae, A. Sato, T. Kurata, L. Jeni. *Training-free Multi-view 4D Human Motion Reconstruction Virtual Reality System*. **WACV** 2026, **CVPR-W** 2026
[4] G. Cazenavette, **J. Julin**, S. Lucey. *Rethinking the Role of Spatial Mixing*. **CVPR-W** 2025
[5] L. Xie, **J. Julin**, K. Niinuma, L. Jeni. *Gaussian Splatting Lucas-Kanade*. **ICLR** 2025
[6] H. Yu, **J. Julin**, Z. Milacski, K. Niinuma, L. Jeni. *CoGS: Controllable Gaussian Splatting*. **CVPR** 2024
[7] H. Yu, **J. Julin**, Z. Milacski, K. Niinuma, L. Jeni. *DyLiN: Making Light Field Networks Dynamic*. **CVPR** 2023
[8] Y. Jang, et al. *VSCHH 2023: A Benchmark for the View Synthesis Challenge of Human Heads*. **ICCV** 2023
[9] **J. Julin**, H. Yu, L. Jeni. *Automated Content Editing in NeRFs*. **RISS Working Paper Journal** 2022

## U.S. PATENTS

- H. Yu, **J. Julin**, Z. Milacski, K. Niinuma, L. Jeni. *Anatomically Correct Neural Avatars*. **pending**

## TEACHING EXPERIENCE

**Teaching Assistant**
*Learning for 3D Vision (16-825), Carnegie Mellon University* *Fall 2026*

**Teaching Assistant**
*MSCV Capstone (16-621), Carnegie Mellon University* *Fall 2025*

## SERVICE

- Member, 2026 M.S. in Robotics Admissions Committee, Carnegie Mellon University.
- Organizer, 2nd Workshop on Benchmarking Evidence-Aligned Multimodal Reasoning (BEAM 2), **ECCV** 2026
- Reviewer, **CVPR** 2026, **ECCV** 2026, **NeurIPS** 2026

## TALKS & PRESENTATIONS

- *Training-free Multi-view 4D Human Motion Reconstruction Virtual Reality System*, oral presentation, SPAR-3D Workshop, **CVPR** 2026

## AWARDS

- **Best Paper Award**, SPAR-3D Workshop, **CVPR** 2026
- 8th China International "Internet +" Competition, one of 2 international projects selected from 3.4 M submissions across 111 countries, **Gold Medal**
- Funded by NSF from May 2022 to Aug. 2022.
- HackPSU Capital One Challenge. **1st Place**

## TECHNICAL SKILLS

**Programming Languages:** Python, C, Java, HTML
**Libraries and Tools:** PyTorch, Tensorflow, OpenCV, Blender, Unity, COLMAP
**Methods and Techniques:** 3D Gaussian Splatting, Neural Radiance Fields (NeRF), Dynamic Scene Reconstruction, Scene Flow Estimation, Neural Rendering

## RELEVANT COURSEWORK

Computer Vision, Learning for 3D Vision, Math Fundamentals for Robotics, Visual Learning and Recognition, Learning-based Image Synthesis, Mobile Robots, Robot Learning